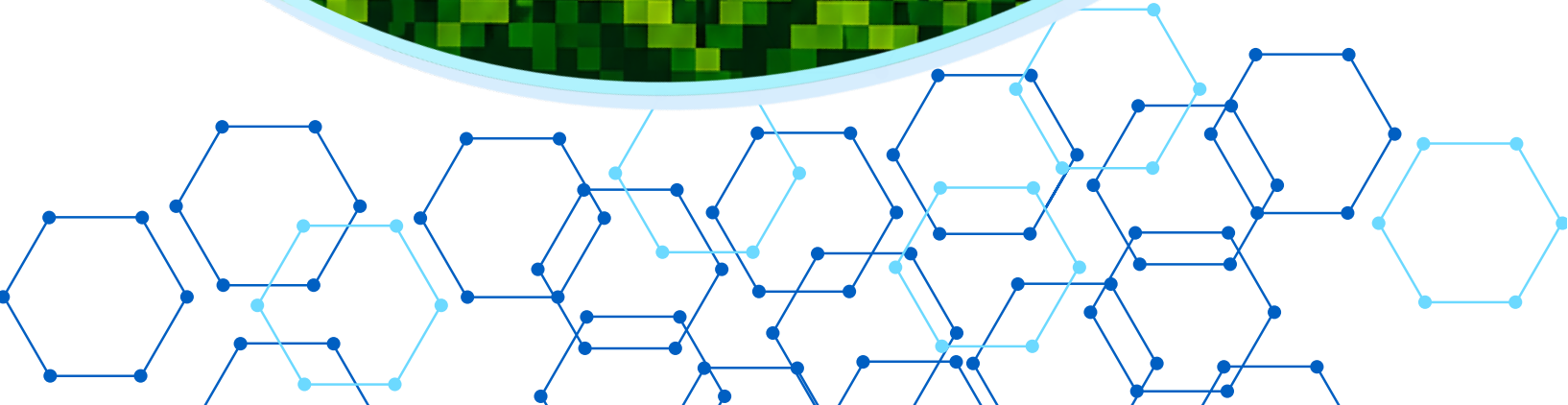


Presented by:



UNI TEXTBOOK MARKETPLACE



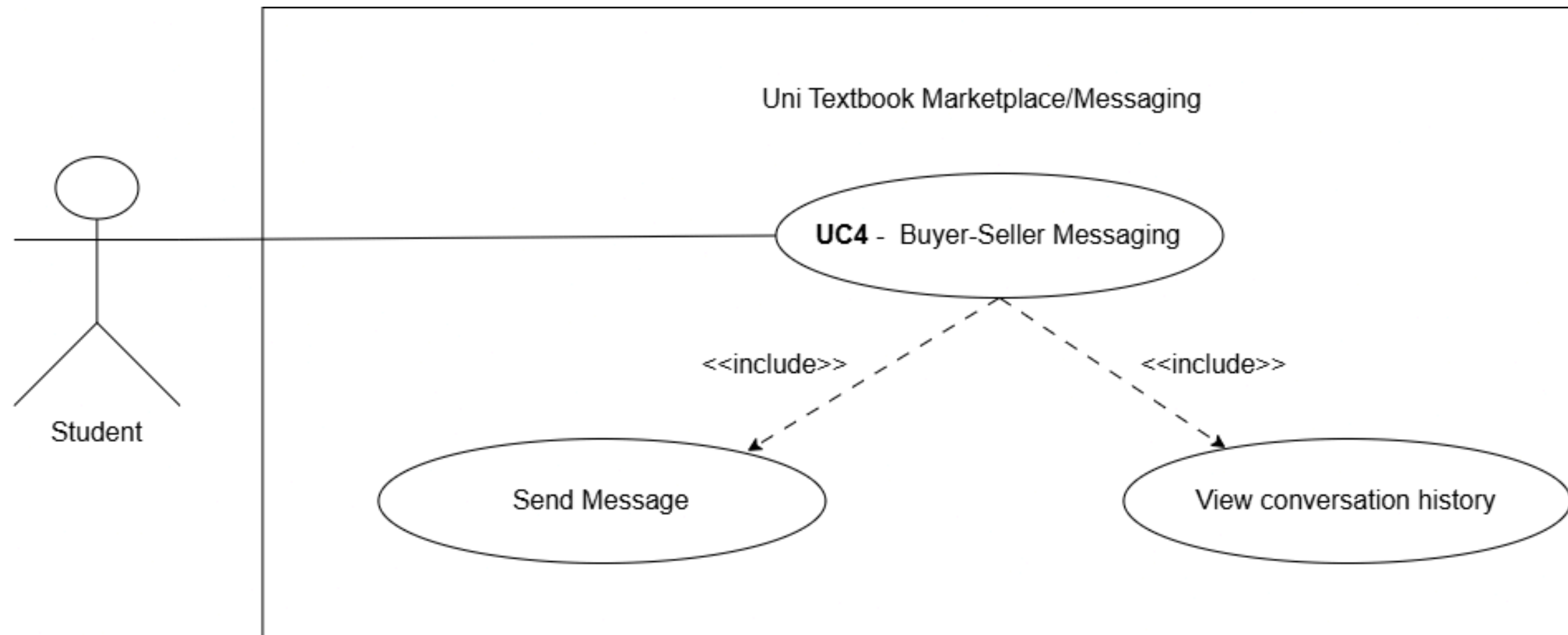
THE UNI-TEXTBOOK MARKETPLACE IN A NUTSHELL

Students struggle to buy/sell textbooks on WhatsApp and Facebook. Listings lack details. Sellers are unverified. Scams are common.



A university-verified marketplace with structured listings, module browsing, smart filters, and privacy-first messaging.

IMPLEMENTED USE CASES

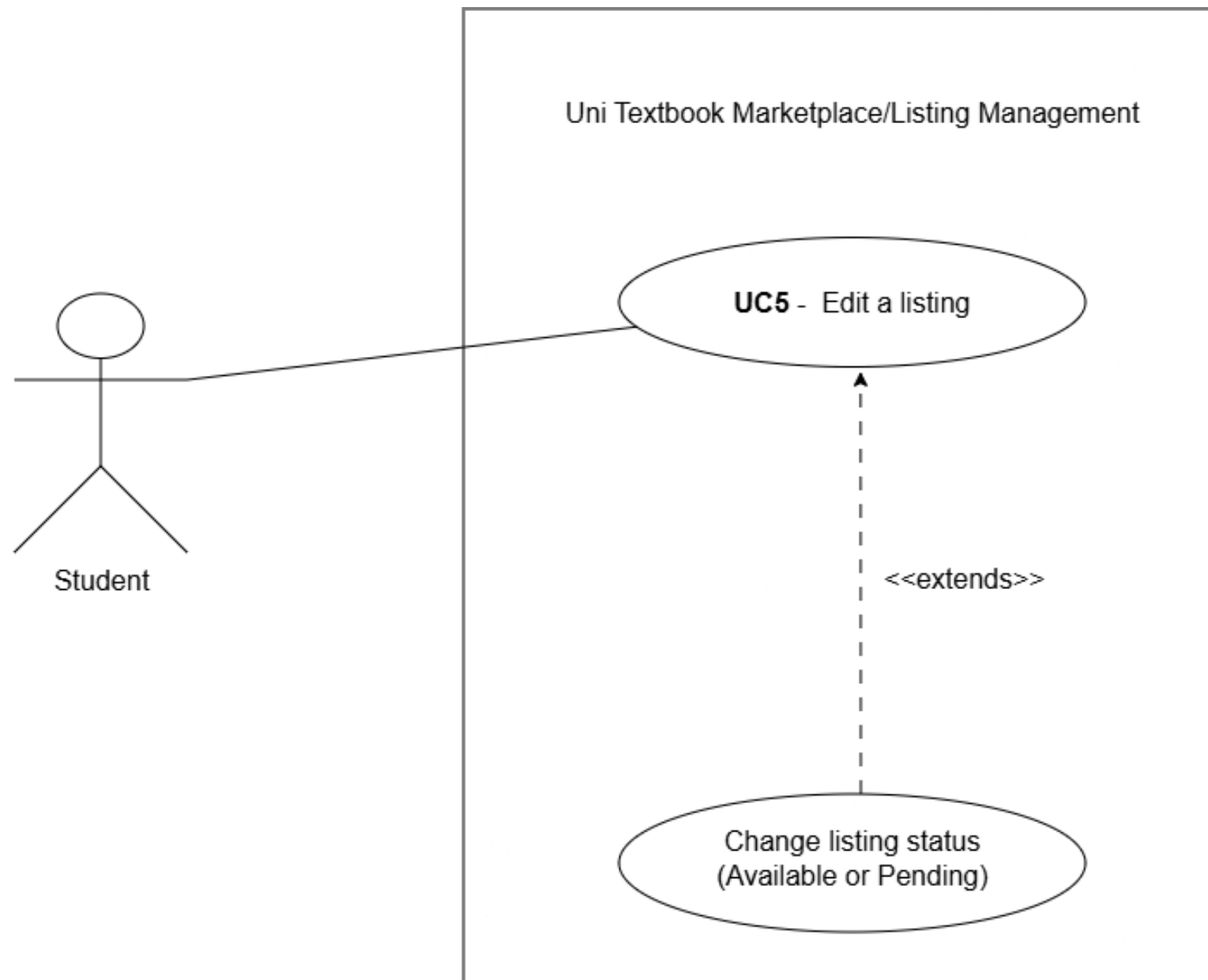


UC4: IN-APP MESSAGING

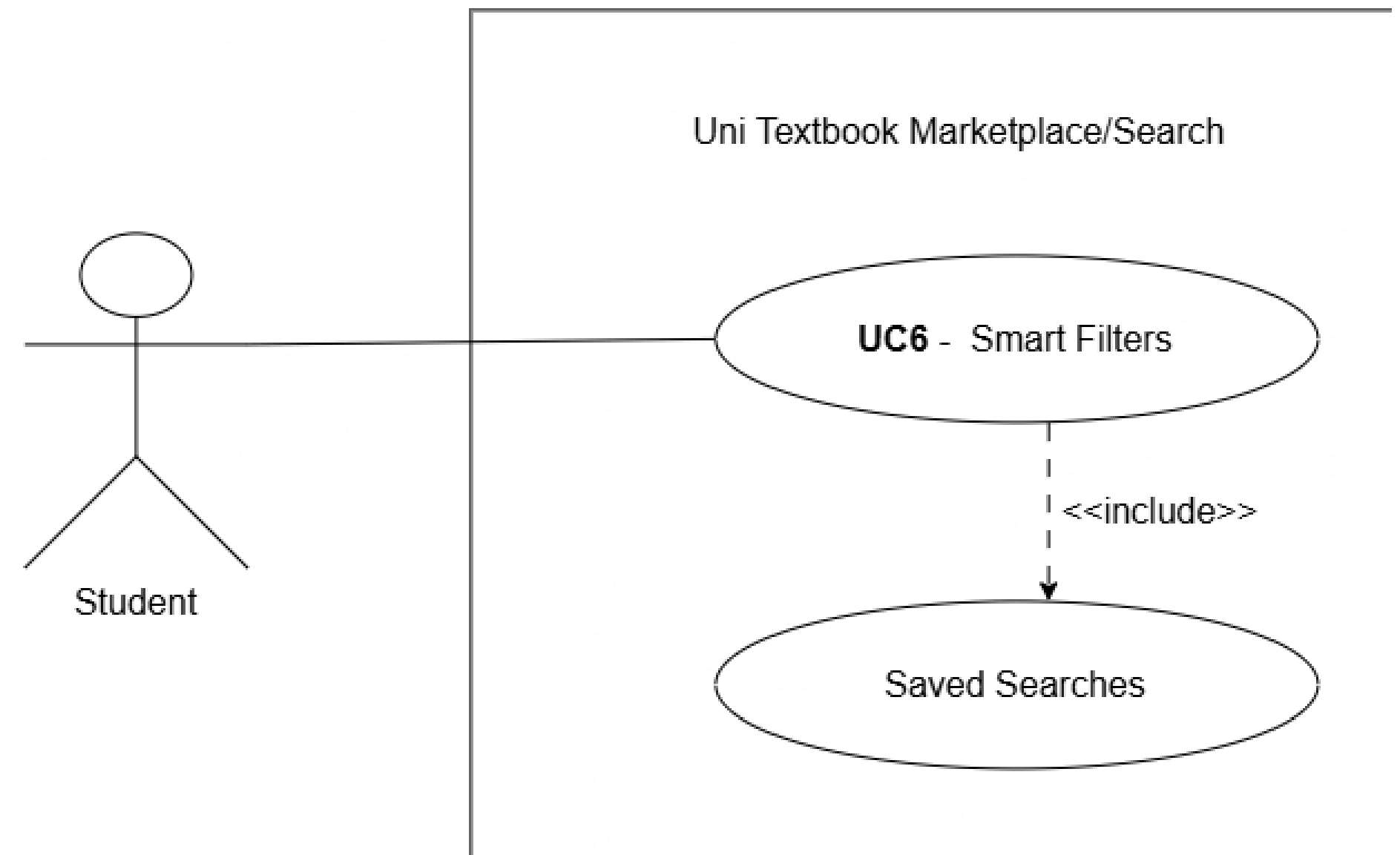


IMPLEMENTED USE CASES

UC5: EDITING A LISTING



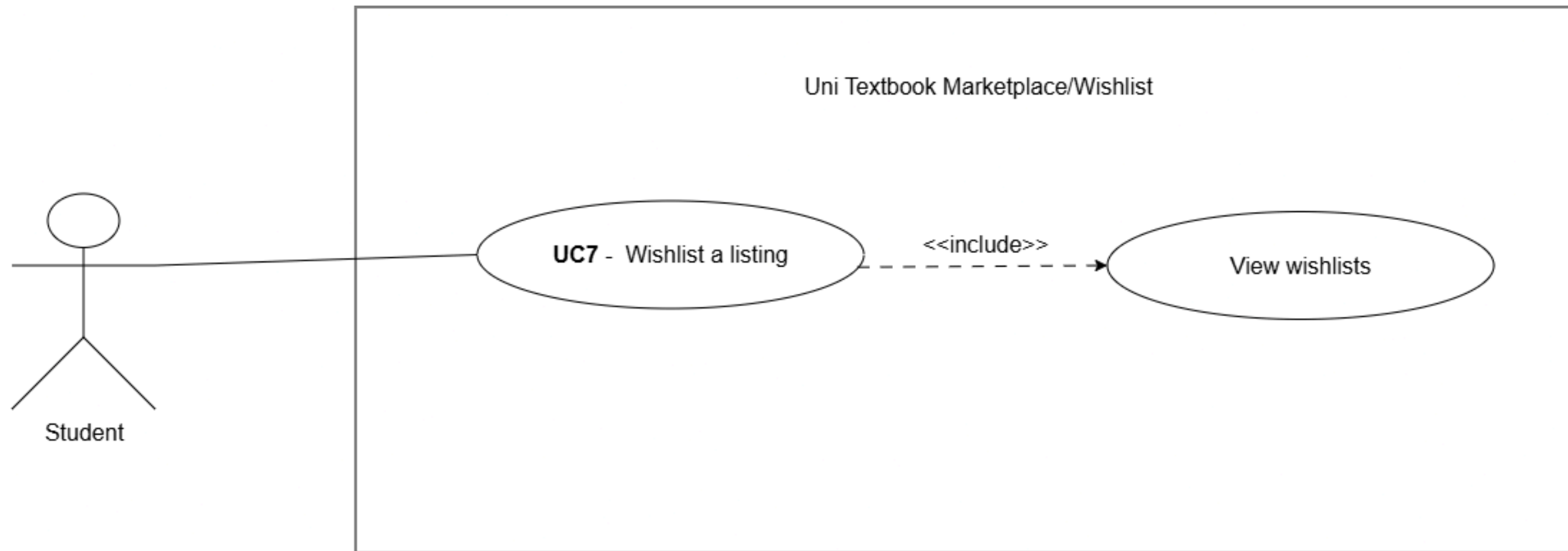
UC6: SMART FILTERS & SAVED SEARCHES



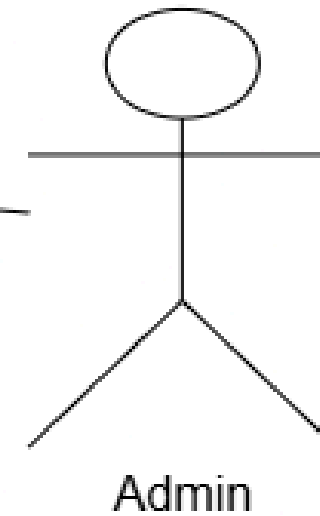
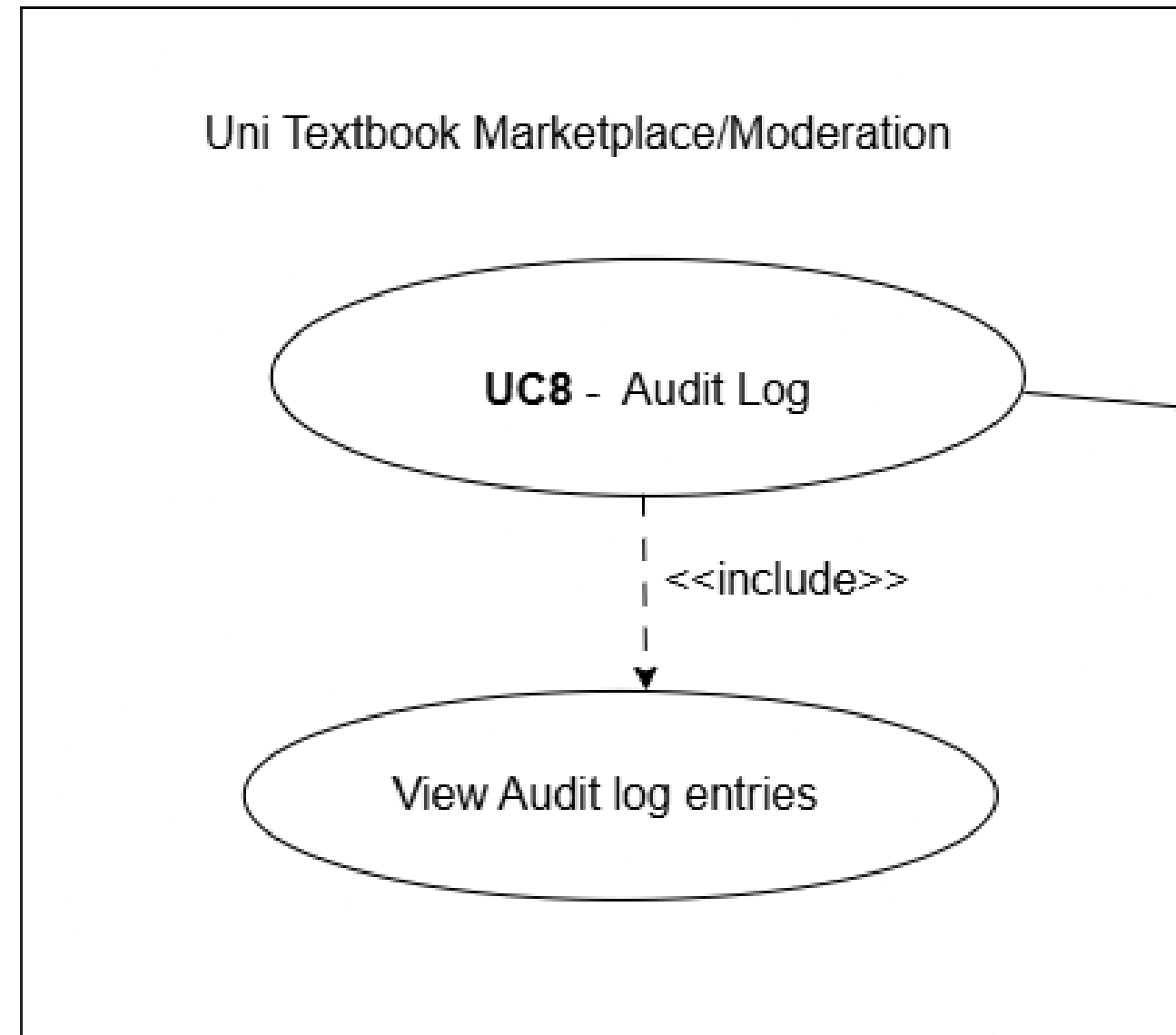
IMPLEMENTED USE CASES

UC7: WISHLIST

UC8: AUDIT LOG QUERY



IMPLEMENTED USE CASES



**UC8: AUDIT LOG
QUERY**



CI/CD PIPELINE



Push to main/develop

ci.yml - Quality Gate

Backend Job

PsqL container

npm lint +
npm test:cov

Frontend Job

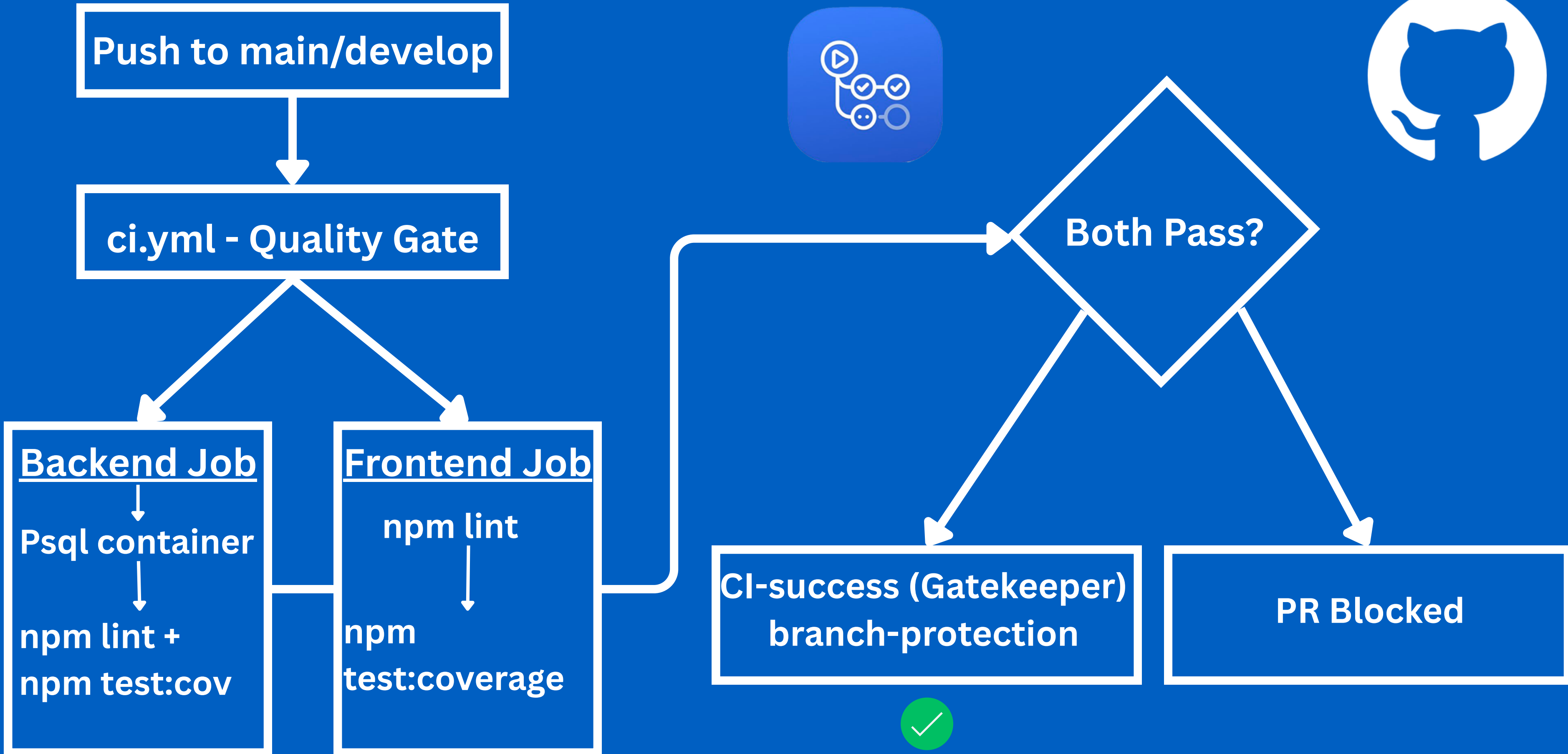
npm lint

npm
test:coverage

Both Pass?

CI-success (Gatekeeper)
branch-protection

PR Blocked



DEPLOYMENT



Push to main/develop

deploy.yml - Build & Deploy

Deploy backend

↓
az acr build,
tag = commit
SHA
↓
az
containerapp
update

Deploy frontend

az acr build --
build-arg
NEXT_PUBLIC_
API_URL
↓
az containerapp
update

Branch?

<< main >>

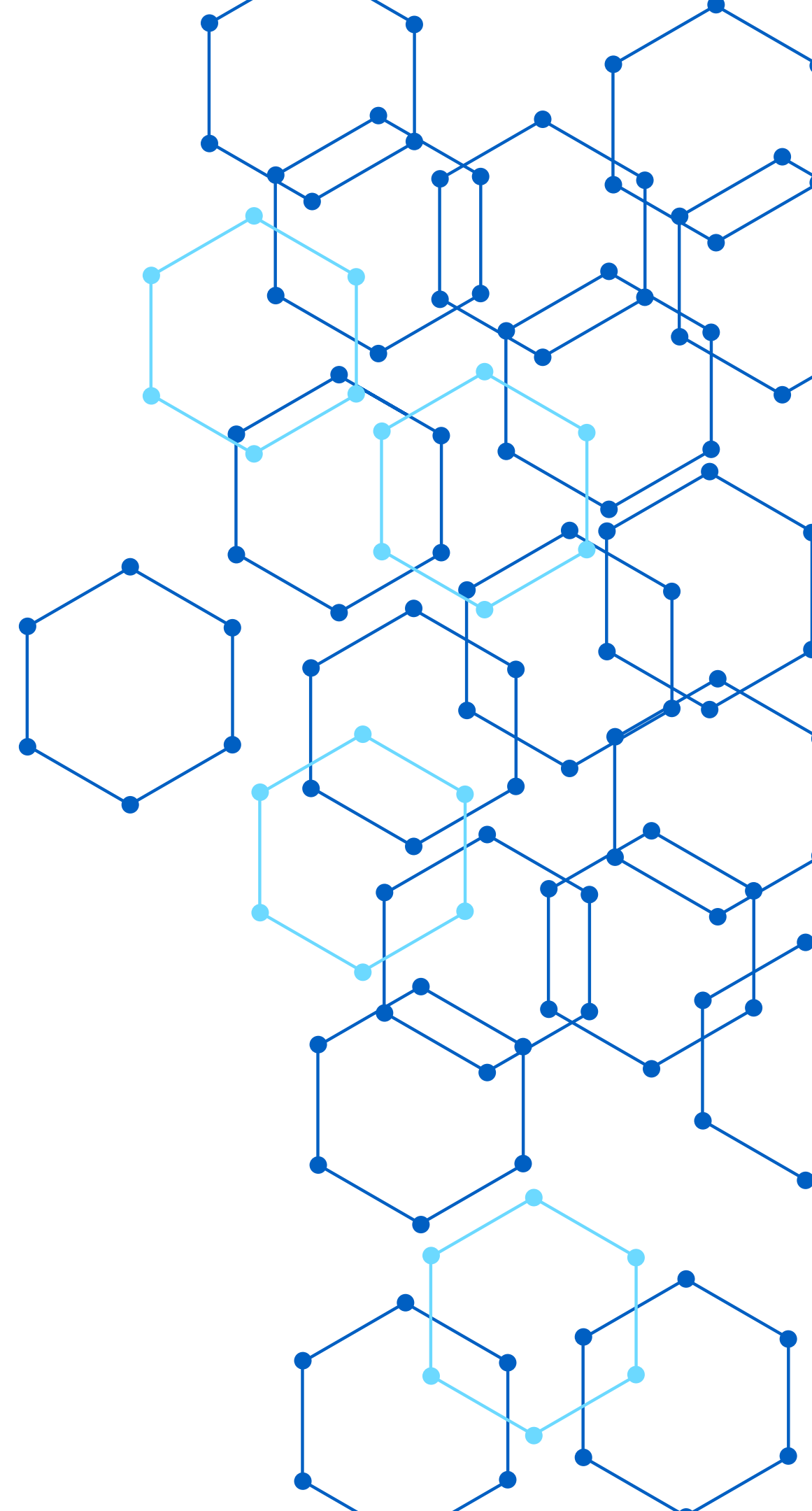
<< develop >>

Production
LIVE

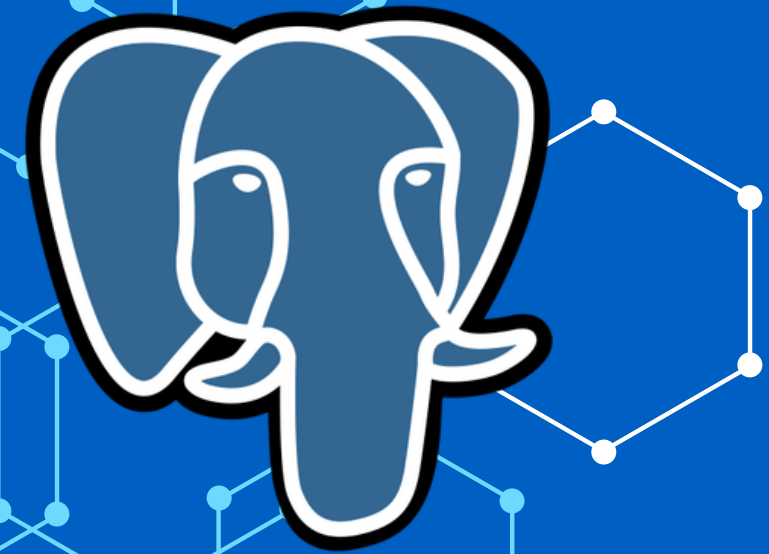
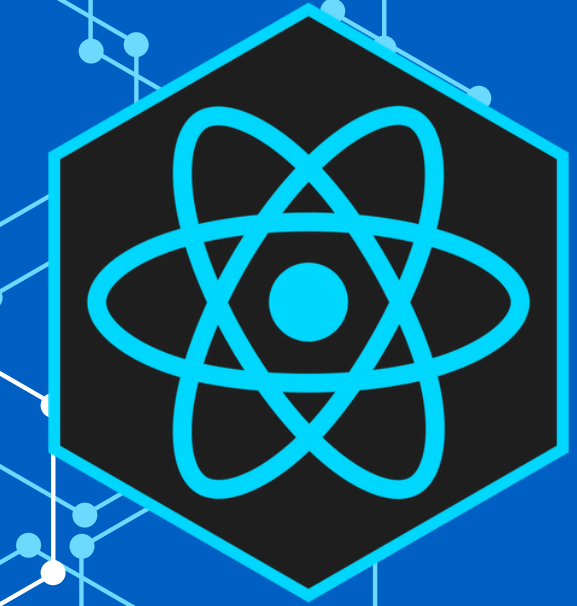
Staging



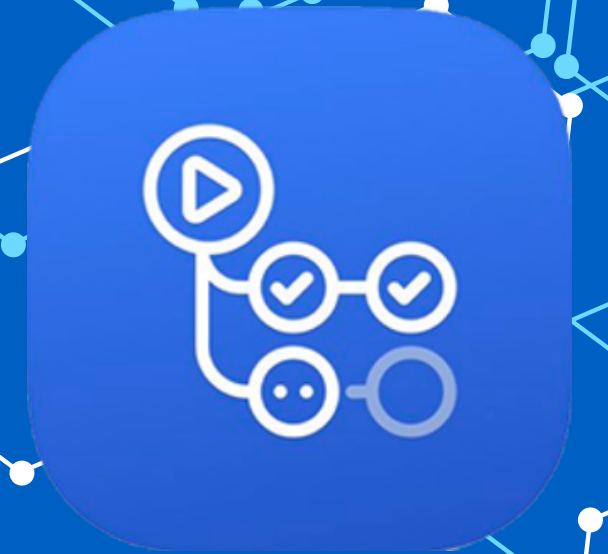
LIVE DEMO



TECH STACK



COMPONENT	TECHNOLOGY
Frontend	NextJS + Tailwind CSS
Backend	NestJS
Database	PostgreSQL
In-app messaging	Firebase
Auth	JWT
CI/CD	GitHub Actions
Host	MS Azure



INTEGRATION TESTS

```
RolesGuard: Checking roles...
```

```
at RolesGuard.canActivate (src/auth/guards/roles.guard.ts:25:13)
```

```
PASS test/admin.integration.spec.ts (215.449 s)
```

```
Admin Integration Tests
```

```
Database Connection
```

```
✓ should load app and connect to database (7758 ms)
```

```
Admin Approve Listing
```

```
✓ should approve a pending listing and create audit log (18633 ms)
```

```
✓ should return 403 when non-admin tries to approve (11468 ms)
```

```
✓ should return 404 when listing not found (11337 ms)
```

```
Admin Reject Listing
```

```
✓ should reject a pending listing and create audit log (15642 ms)
```

```
✓ should return 403 when non-admin tries to reject (11709 ms)
```

```
Get Audit Logs
```

```
✓ should retrieve audit logs for admin (15155 ms)
```

```
✓ should filter audit logs by action (14917 ms)
```

```
✓ should return 403 when non-admin tries to view audit logs (11398 ms)
```

```
Get Admin Users
```

```
✓ should retrieve all admin users (11228 ms)
```

```
✓ should return 403 when non-admin tries to view admin users (11185 ms)
```

```
Complete Admin Workflow
```

```
✓ should handle complete workflow: create listing → approve → audit log → view admin users (15137 ms)
```

```
Test Suites: 1 passed, 1 total
```

```
Tests: 12 passed, 12 total
```

```
Snapshots: 0 total
```

```
Time: 218.466 s, estimated 317 s
```

```
Ran all test suites matching /admin.integration.spec.ts/i.
```

UNIT TESTS

```
aging/**/*.controller.ts" --collectCoverageFrom="!src/messaging/dto/**/*.ts" test/messaging.service.spec.ts
```

PASS test/messaging.service.spec.ts

MessagingService

createConversation

- ✓ throws NotFoundException when the listing does not exist (19 ms)
- ✓ throws ForbiddenException when the buyer is the seller (2 ms)
- ✓ returns the existing conversation when one already exists (2 ms)
- ✓ creates a new conversation when none exists (3 ms)

getMyConversations

- ✓ maps the seller straight from the listing when they're the other party (2 ms)
- ✓ falls back to the users repository when the other party is not the seller (1 ms)
- ✓ skips conversations whose listing no longer exists (1 ms)
- ✓ skips conversations whose other participant no longer exists (2 ms)
- ✓ sorts conversations by updatedAt, most recent first (2 ms)

getMessages

- ✓ throws NotFoundException when the conversation does not exist (2 ms)
- ✓ throws ForbiddenException when the user is not a participant (1 ms)
- ✓ returns messages ordered oldest to newest (1 ms)

sendMessage

- ✓ throws NotFoundException when the conversation does not exist (1 ms)
- ✓ throws ForbiddenException when the sender is not a participant (1 ms)
- ✓ adds the message and updates the conversation summary (1 ms)

File	% Stmts	% Branch	% Funcs	% Lines	Uncovered Line #s
All files	100	90.47	100	100	
messaging.service.ts	100	90.47	100	100	185-189

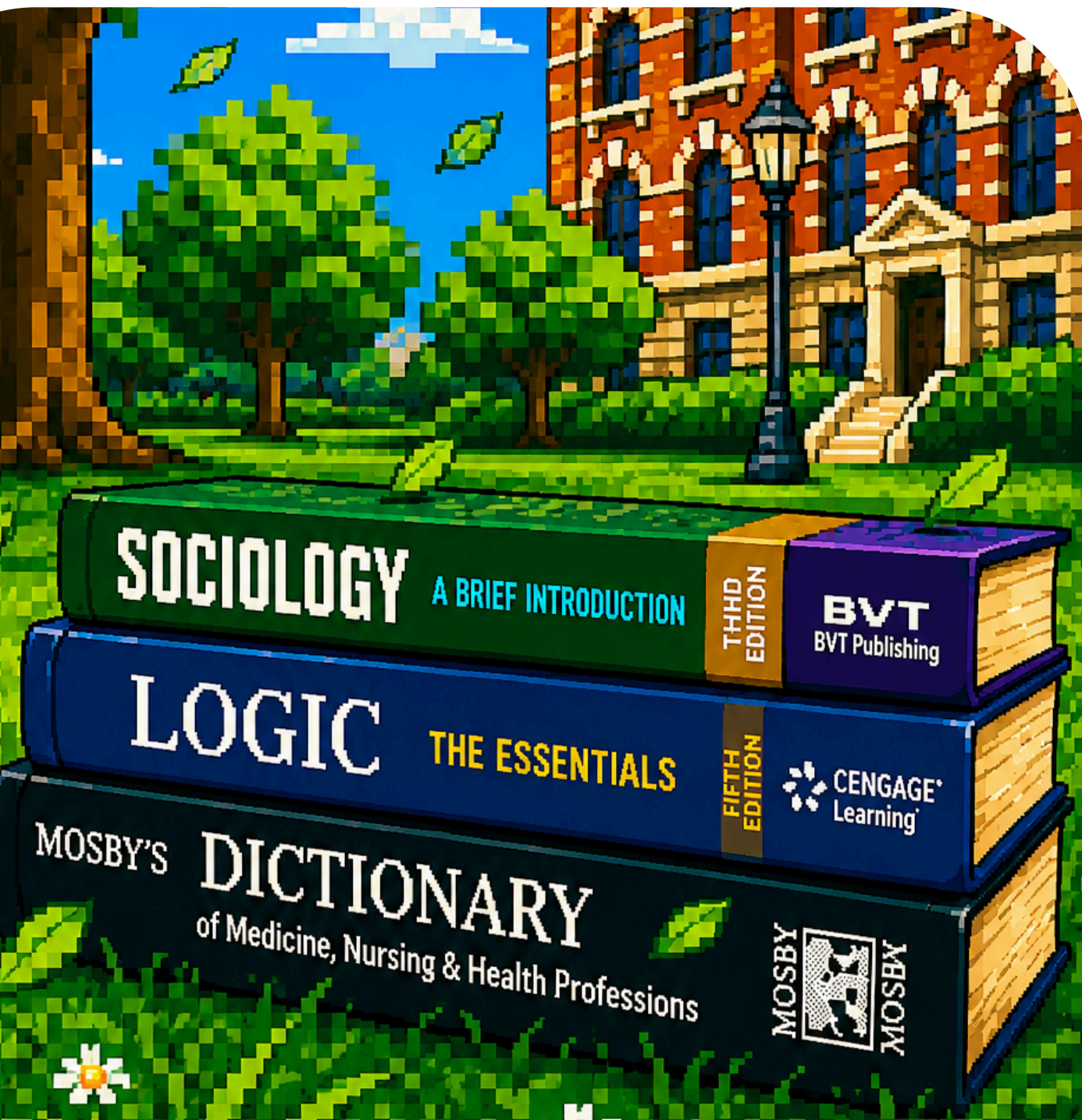
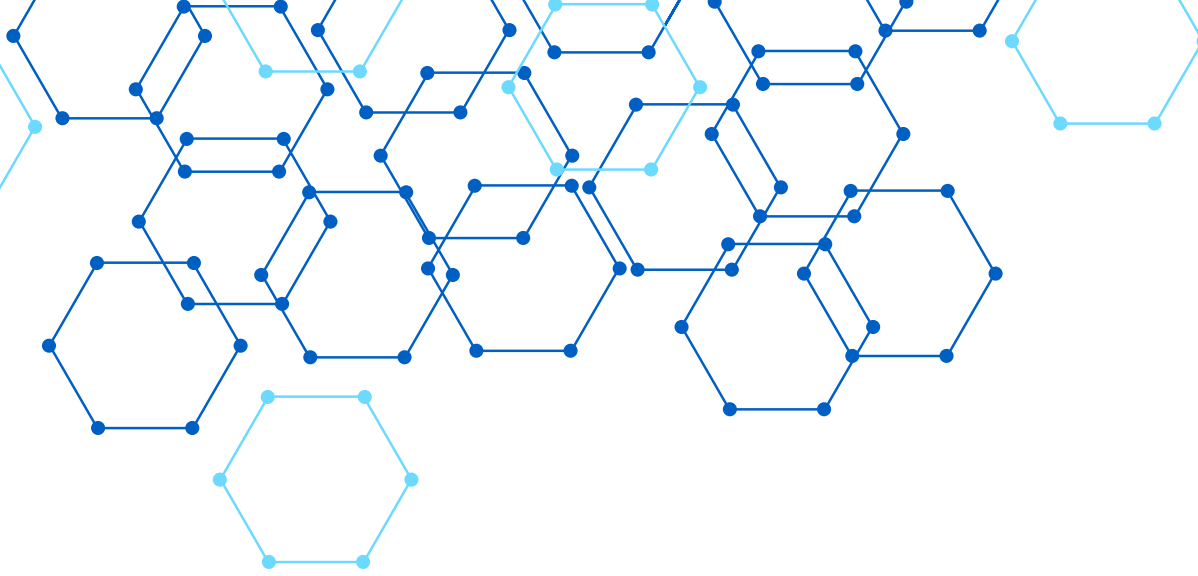
Test Suites: 1 passed, 1 total

Tests: 15 passed, 15 total

Snapshots: 0 total

Time: 4.196 s

Ran all test suites matching /test\/messaging.service.spec.ts/i.



THANK YOU

agilebridge
CONSIDERED
SOFTWARE SOLUTIONS
Capstone 2026

